INCH-POUND

MIL-F-63446A (AR)
AMENDMENT 8
7 July 1989
SUPERSEDING
AMENDMENT 7
23 August 1985

MILITARY SPECIFICATION

FUZES, GENERAL PURPOSE GRENADE:
M230 AND M231 (PRACTICE)

This Amendment forms a part of Military Specification MIL-F-63446A (AR), dated 12 September 1983, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

Page 2

2.1.2, Delete "Fuze Assembly-M230 All ways" and substitute "Fuze General Purpose Grenade. M230".

Delete "Fuze Assembly-M231 All Ways" and substitute "Fuze, General Purpose Grenade, Practice: M231".

Page 3

3.4.1, Delete in its entirety and substitute:

"3.4.1. Molded and Plastic Parts. Dimensions on plastic parts that are a function of core and cavity sizes of the mold itself shall be inspected at reduced levels. One tenth of one percent (0.1%) of a production lot shall be inspected per cavity. One dimension across the parting line shall be inspected per the requirements of General Inspection Level II of MIL-STD-105. A single reject resulting from the reduced inspection of molded dimensions shall cause that cavity to be removed from production until corrective action is taken and the part from that cavity re-qualified. Machined dimensions shall be inspected to Level II of MIL-STD-105."

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3.4.3, Delete in its entirety and substitute:

"3.4.3. Porosity Check. The die cast Fuze Body shall be inspected for porosity in accordance with 4.5.1 of this specification."

3.4.4.2, Delete in its entirety and substitute:

"3.4.4.2 Spring Set. The Slider spring and firing pin spring shall maintain drawing load requirements after being compressed at elevated temperatures".

3.4.6, Delete: ".125 inch in diameter."

PAGE 5

3.5.7, Line 1, Delete: "18 ± 1.5 pounds" and substitute "19 ± 3 pounds".

Line 2, Delete: "a pull load" and substitute "a constant pull rate."

3.5.8, Delete: "two pounds" and substitute "eight pounds".

Add new sentence to end of paragraph "The load shall be applied at a constant rate."

3.5.9, Delete: "1.75 ± .25 seconds" and substitute "1.50 ± .50 seconds."

3.5.12, Delete in its entirety.

PAGE 6

Add new paragraphs 3.8 and 3.9:

"3.8 Arming pin/slider lock. The arming pin shall lock the slider in its fully safe position when the fuze is fully assembled."

"3.9 Auto trigger. The trigger shall not function spontaneously when the slider is moved to the fully armed position."

PAGE 7

4.3.1, Delete: "Fuze Subassembly No.assembly No. 1", "Fuze Subassembly No. 2", "Fuze Subassembly No. 4", "Fuze Subassembly No: 5", and "Fuze Subassembly No: 6" in their entirety.

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Delete: "Slider 9333807 50/Cavity" and substitute "Slider 9333807 25/Cavity".

Delete: "Body Fuze 9333824 50/Cavity" and substitute "Body, Fuze 9333824 25/Cavity".

Delete: "Spring Slider 9333811 50/ and substitute "Spring, Slider 9333811 25/Cavity".

Delete: "Fuze Subassembly No: 3" and substitute "Fuze Body Subassembly"

Page 8

4.3.1, Delete: "Fuze Assembly, M231 Allways" and substitute "Fuze, General Purpose Grenade, Practice: M231".

Delete: "Fuze Assembly, M230 Allways" and substitute "Fuze General Purpose Grenade: M230".

Add: under Submisssion "Drive Screw, Modified 9390620 25".

Add: "Barrier Plate Assembly with Retainer Tape 12561493 50"

Page 9

TABLE I: Delete "50" and substitute "25" in four places.

PAGE 10

TABLE I: Delete "Fuze Subassembly No. 1" and "Fuze Subassembly No. 2" in their entirety.

Delete "Fuze Subassembly No. 3" and substitute "Fuze Body Subassembly".

Delete "Pushout requirement" and substitute "Anti-twist pin pushout requirement".

Add: "sleeve pushout requirement 25 3.4 Gage".

Page 11

TABLE I: Delete "Fuze Subassembly No. 4" in its entirety.

PAGE 15

TABLE I: Delete "Fuze Subassembly No. 5" in its entirety.

PAGE 16

TABLE I: Delete "Fuze Subassembly No. 6" in its entirety.

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PAGE 17

TABLE I: Add under Plate Assembly Barrier "Lead Pull test 25 3.4 4.4.2.49"

Add: "Drive Screw, modified
(Dwg. 9390620)
Examination for Defects 25 3.4 4.4.2.52"

Add: "Barrier Plate Assembly with Retainer Tape
(Dwg. 12561493)
Examination for defects 50 3.4 4.4.2.53"

PAGE 18

TABLE I: Delete "M231 All Ways Fuze Assembly" and substitute "Fuze, General Purpose Grenade, Practice: M231".

Add:

"Auto Trigger	115 (a)	3.9	4.5.15 (b)
Torque Requirement	115 (a)	3.4	4.4.2.50

Delete: "239" and substitute "239 (a)."

Delete: "Transportation Vibration 20 3.5.3 4.5.8/4.5.15/4.5.16" and substitute "Transportation Vibration 20 3.5.3 4.5.8/4.5.15 (c)/4.5.16 (c),.

Delete: "Helicopter Vibration 30 3.5.4 4.5.9/4.5.15/4.5.16" and substitute "Helicopter Vibration 30 3.5.4 4.5.9/4.5.15 (c)/4.5.16 (c)".

Delete: "Thermal Shock 30 3.5.6 4.5.11/4.5.15/4.5.16" and substitute "Thermal Shock 30 3.5.6 4.5.11/ 4.5.15 (c)/4.5.16 (c)".

Delete: "Arming Time 115 (a) 3.5.9 4.5.14" and substitute "Arming Time 115(a) 3.5.9 4.5.14(f)".

Delete: "Fuzing No-Go 115 (a) 3.5.10 4.5.15 (b)" and substitute "Fuzing No-Go 49 (d) 3.5.10 4.5.15 (b)".

Delete: "Fuzing Go 115 (a) 3.5.11 4.5.16 (b)" and substitute "Fuzing Go 66 (d) 3.5.11 4.5.16 (e)".

Delete: note (a) in its entirety and substitute:

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(a). The same fuzes are to be subjected to this test. However, if the test within 90 second requirement cannot be maintained, each test or any combination of these tests may be performed on a separate group of fuzes which could increase the maximum sample size to 584 fuzes."

In Note (b), Delete "32" and substitute "16" in two places; Delete "51" and substitute "17".

Add to the Notes Block:

"(c) Fuzing No-Go - 1/2 test sample per temperature; Fuzing Go - 1/2 test sample per temperature.

(d) Selected from fuzes marked (a).

(e) Prior to performing tests, condition fuzes for two hours after stabilization at indicated temperatures; $-50^{\circ} \pm 5^{\circ}$F -22 Fuzes; 70° F $\pm 15^{\circ}$F -22 Fuzes; 160°F $\pm 5^{\circ}$F -22 Fuzes. The Fuzing Go and No-Go impulse shall be imparted to the fuze within 90 seconds of removal from temperature conditioning chamber. All fuzes tested at cold shall be conditioned while sealed in plastic bags.

(f) Tests to be performed at cold, ambient, and hot temperatures in quantities described in (b) and (e)."

PAGE 19

TABLE I: Delete "M230 All Ways Fuze Assembly" and substitute "Fuze, General Purpose Grenade, M230".

Add

"Torque Requirement	115 (a)	3.4	4.4.2.51
Auto Trigger	115 (a)	3.9	4.5.15 (b)"

Delete: "239" and substitute "239 (a)."

Delete: "Transportation Vibration 20 3.5.3 4.5.8/4.5.15 4.5.16" and substitute "Transportation Vibration 20 3.5.3 4.5.8/4.5.15 (c)/4.5.16 (c)".

Delete: "Helicopter Vibration 30 3.5.4 4.5.9/4.5.15/ 4.5.16" and substitute "Helicopter Vibration 30 3.5.4 4.5.9/4.5.15 (c)/4.5.16 (c)".

Delete: "Thermal Shock 30 3.5.6 4.5.11/4.5.15/4.5.16" and substitute "Thermal Shock 30 3.5.6 4.5.11/4.5.15 (c)/4.5.16 (c)".

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:53Z
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Delete: "Arming Time 115 (a) 3.5.9 4.5.14 (b)" and substitute "Arming Time 115 (a) 3.5.9 4.5.14 (f)".

Delete: "Fuzing No-Go 115 (a) 3.5.10 4.5.15 (b)" and substitute "Fuzing No-Go 49 (d) 3.5.10 4.5.15 (b)".

Delete: "Fuzing Go 115 (a) 3.5.11 4.5.16 (b)" and substitute Fuzing Go 66 (d) 3.5.11 4.5.16 (e)".

Delete note (a) in its entirety and substitute:

"(a). The same fuzes are to be subjected to these tests. However, if the test within 90 second requirement cannot be maintained, each test or any combination of these tests may be performed on a separate group of fuzes which could increase the maximum sample size to 584 fuzes."

In Note (b), Delete "32" and substitute "16" in two places; Delete "51" and substitute "17".

Add to the Notes Block:

(c) Fuzing No-Go - 1/2 test sample per temperature; Fuzing Go - 1/2 test sample per temperature.

(d) Selected from fuzes marked (a).

(e) Prior to performing tests, codition fuzes for two hours after stabilization at indicated temperatures; $-50^{\circ}F \pm 5^{\circ}F$ -22 Fuzes; $70^{\circ} F \pm 15^{\circ}F$ -22 Fuzes; $160^{\circ}F \pm 5^{\circ}F$ -22 Fuzes. The Fuzing Go and Fuzing No Go impulse shall be imparted to the fuze within 90 seconds of removal from temperature conditioning chamber. All fuzes tested at cold shall be conditioned while sealed in plastic bags.

(f) Tests to be performed at cold, ambient, and hot temperatures in the quantities described in (b) and (e)."

Page 21

4.4.2.1, Critical: Delete "None defined".

Add Critical 1: "1 Presence of red lacquer on bore rider 100% 3.4 Visual".

Major 102: Delete "width of slider" and substitute "width of slider rear".

Major 103: Delete "thickness of slider" and substitute "thickness of slider rear".

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Add "(a)" next to .40% in sixteen places.

Add note "(a) See paragraph 3.4.1 for applicability of reduced inspection."

Add majors 117, 118 and 119:

"117: Minimum coverage of red lacquer on bore rider .40% (a) 3.2 Gage".

"118: Width of bore rider tip .40% (a) 3.4 Gage".

"119: Thickness of slider front .40% (a) 3.4 Gage".

Page 22

4.4.2.1, Add "(a)" next to .65% in eight places.

Add note "(a) See paragraph 3.4.1 for applicability of reduced inspection."

Add minors 209, 210, 211 as follows:

"209 Maximum coverage of red lacquer on bore rider. .65% (a) 3.4 Gage.

210 Width of slider between ends .65% (a) 3.4 Gage.

211 Thickness of slider between ends .65% (a) 3.4 Gage".

Page 23

4.4.2.2, Major 101, add "(b)" under no. of sample units.

Delete Majors 102, 103, 109, 113, 115 in their entirety.

Delete note (a) in its entirety and substitute:

"(a). reduced sampling based on QQ-591F. Sample size to equal a minimum of 0.1% of a production lot per cavity for cast dimensions only: Machined dimensions shall be inspected to General Inspection Level II of MIL-STD-105, .40% for majors and .65% for minors: For reduced sampling, accept/reject is 0/1."

Add note (b) "(b): See TABLE II of QQ-591F for accept/reject numbers."

Page 24

4.4.2.2, Add Major 129:

Source: https://assist.dla.mil -- Downloaded: 202610-23T02:53Z
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"129 Diameter of Drive Screw Holes .40% 3.4 Gage

Delete Majors 118, 119, 120, 122, 124, 125, 127, 128 in their entirety.

Major 121, delete "center" and substitute "center side of slider slot".

Delete Minor 201 in its entirety and substitute "201 height of firing pin bore (a) 3.4.2 Gage".

Delete Minor 206 in its entirety and substitute "206 Diameter of firing pin bore at slider slot (a) 3.4.2 Gage".

Delete note (a) in its entirety and substitute:

"(a). Reduced sampling based on QQ-591F. Sampling size to equal a minimum of 0.1% of a production lot per cavity for cast dimensions only. Machined dimensions shall be inspected in accordance with General Inspection Level II of MIL-STD-105, .40% for majors and .65% for minors. For reduced sampling, accept/reject is 0/1."

4.4.2.2, Delete Minor 209 in its entirety and substitute "209 Depth of bore rider slot (a) 3.4.2 Gage":

Add Minors 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, and 221 as follows:

"211 Max. depth of relief surface	(a)	3.4.2	Gage
212 Diameter, center hole	(a)	3.4	Gage
213 Diameter of anti-twist pin holes	(a)	3.4	Gage
214 Distance between anti-twist pin holes	(a)	3.4	Gage
215 Diameter of detent ball hole	(a)	3.4	Gage
216 True position, detent ball hole to pivot thread	(a)	3.4	Gage
217 Thickness of flange	(a)	3.4	Gage
218 Chamfer on flange	(a)	3.4	Gage
219 Diameter of safety pin hole	(a)	3.4	Gage
220 Orientation of notches (First Article only)	(a)	3.4	Gage
221 Diameter, RSP Holes	(a)	3.4	Gage

Delete note (a) in its entirety and substitute:

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"(a) Reduced sampling based on QQ-591F. Sample size to equal a minimum of 0.1% of a production lot per cavity for cast dimensions only. Machined dimensions shall be inspected in accordance with General Inspection Level II of MIL-STD-105, .40% for majors and .65% for minors. For reduced sampling, accept/reject is 0/1."

Page 28

4.4.2.5, Delete in its entirety.

Page 29

4.4.2.6, Major 104, delete "Gage" and substitute "Visual".

Page 30

4.4.2.7, Delete in its entirety.

Page 31

4.4.2.8, Delete Majors 104 and 105 in their entirety.

Delete Minor 201 in its entirety and substitute "201 Body diameter .65% 3.4 Gage".

Delete Minor 202 in its entirety and substitute "202 True position, I.D. to O.D. .65% 3.4 Gage".

Delete Minors 203, 204, 205 in their entirety.

Page 32

4.4.2.9, Delete "Fuze Subassembly No. 3" and substitute "Fuze Body Assembly".

Delete "9347186" and substitute "9333825 or 9345168".

Add Majors 102, 103, 104 and 105 as follows:

"102 Pins above bottom of slider slot .40% 3.4 Gage
103 Anti-twist pin pushout requirement .40% 3.4 Gage
104 Adhesive in not-called-for area .40% 3.4 Visual
105 Wrong adhesive 100% 3.4 Certification"

Delete Minors "201", "202", and "203" in their entirety.

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Minor 204: Delete "Pushout requirement" and substitute "Sleeve pushout requirement"

Page 33

4.4.2.10: Delete in its entirety.

Page 34

4.4.2.11, Add Major 111:

"111 Finish on tip (a) 3.4 Gage"

Delete Majors 101, 105, 107, 109 in their entirety..

Delete Minor 201 in its entirety and substitute "201 Length of Major O.D. .65% 3.4 Gage"

Add Minors 203 and 204 as follows:

"203 Relieved diameter .65% 3.4 Gage
204 True position of pivoting diameter to major body diameter .65% 3.4 Gage"

Add to Notes block:

"(a) Special Inspection Level 4 of MIL-STD-105D"

Page 35

4.4.2.12, Change "Critical 1, Location of shear wire hole from flange, 100%, 3.4, Gage" to "Major 111, Location of shear wire hole from flange, .40%, 3.4, Gage"

Delete Major 105 in its entirety and substitute

"105 Tip length .40% 3.4 Gage"

Delete Majors 106, 108, and 109 in their entirety.
Delete Minors 201 and 202 in their entirety.

Add Major 110:

"110 Thread improper .40% 3.4 Gage"

Major 102, delete "Lock spring groove location from back" and substitute "True position of blind holes".

Major 103, delete "Diameter of lock spring groove button" and substitute "Depth of blind holes"

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Page 36

4.4.2.13, Delete Major 101 in its entirety.

Page 37

4.4.2.14, Add Majors 104, 105, and 106:

"104 Wrong adhesive 100% 3.4
Certification
105 Improper cure of adhesive 100% 3.4
Visual/Charts (time and condition)
106 Adhesive in not-called-for area 100% 3.4
Visual"

Delete "100%" and substitute ".40%" in two places.

Page 38

4.4.2.15, Add Major 102:

"102 Length end to center of locking loop .40% 3.4
Gage".

Delete Minors 201, 202 and 203 in their entirety.

Page 55

4.4.2.32: Delete "9347187" and substitute "9333825"

Major 101, delete ".40%" and substitute ".40%(a)"

Delete Major 102 and 103 in their entirety.

Add Majors 105, 106, 107 as follows:

"105 Wrong lubricant 100% 3.4
Certification
106 Lubricant missing .40% 3.4 Visual
107 Lubricant in .40% 3.4 Visual"
not-called-for area

Add note (a) to note block "(a). Special Inspection Level
S-4 of MIL- STD-105"

Page 56

4.4.2.33: Delete in its entirety.

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Page 57

4.4.2.34, Delete Major 103 in its entirety.

Delete Minors 201, and 203 in their entirety.

Page 58

4.4.2.35, Add Major 104:

"104 Tip length .40% 3.4 Gage"

Delete "100%" and substitute ".40%" in three places.

Page 59

4.4.2.36, Major 101: Delete "Tip length" and substitute "Straight pin recess,

Delete "100%" and substitute ".40%"

Add Minor 202:

"202 Flush to above .65% 3.4 Gage".

Page 60

4.4.2.37, Delete Majors 104 and 105 in their entirety.

Delete Minor 201, 202, 203, and 204 in their entirety.

Page 62

4.4.2.39, Major 101: Delete "100%" and substitute ".40%".

Major 102: Delete "wire" and substitute "clip".

Page 64

4.4.2.41: Delete in its entirety.

Page 65

4.4.2.42, Major 101 and 104 delete ".40%" and substitute ".40%(a)"

Delete Minor 201 in its entirety:

Add note (a) to note block "(a): Special Inspection Level S-4 of MIL- STD-105"

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Page 69

4.4.2.46, Delete Major 103 in its entirety:

Page 71

4.4.2.48, Delete Major 102 in its entirety:

Page 72

4.4.2.49, Add new Major 103:

"103 Lead Pull Test 50 3.4 4.5.18"

Page 73

4.4.2.50: Delete "M231 All Ways Fuze Assembly" and substitute "Fuze, General Purpose Grenade, Practice: M231"

Delete Criticals 1 and 2 in their entirety:

Add Critical 9, 10, 11, 12, 13, and 14:

"9	Safety pin missing	100%	3.4	Visual
10	Fuzing No-Go	100%(e)	3.5.10	4.5.15
11	Arming pin/slider lock	100%	3.8	4.5.15
12	Presence of aluminum foil tape	100%	3.4	Visual
13	Auto Trigger	100%	3.9	4.5.15
14	Presence of shear wire	100%	3.4	Visual"

Delete Major 104 in its entirety:

Major 105 delete "(a)" and substitute "(d)"

Add Majors 107, 108, 109, and 110 as follows:

"107	Torque requirement	115	3.4	Gage
108	Wrong adhesive	100%	3.4	Certification
109	Improper cure of adhesive (time and condition)	100%	3.4	Visual/Charts
110	Excessive adhesive (internal)	15	3.4	4.5.18"

Delete "Transportation Vibration 12 3.5.3 4.5.8/4.5.15 4.5.16" and substitute "Transportation Vibration 12 3.5.3 4.5.8/4.5.15 (c)/4.5.16 (c)"

Delete "Helicopter Vibration 12 3.5.4 4.5.9/4.5.15/4.5.16" and substitute "Helicopter Vibration 12 3.5.4 4.5.9/4.5.15 (c)/4.5.16 (c)"

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Delete "Thermal Shock 12 3.5.6 4.5.11/4.5.15/4.5.16" and substitute "Thermal Shock 12 3.5.6 4.5.11/4.5.15 (c)/4.5.16 (c)"

Delete "Shear Wire Pull Test 32 (a) 3.5.7 4.5.12" and substitute "Shear Wire Pull Test 64 (a) 3.5.7 4.5.12(f)"

Delete "Safety Pin Pull Test 32 (a) 3.5.7 4.5.12" and substitute "Safety Pin Pull Test 64 (a) 3.5.7 4.5.12(f)"

Delete "Arming Time 32(a) 3.59 4.5.14(b)" and substitute "Arming Time 64(a) 3.59 4.5.14(f)"

Delete note (a) in its entirety and substitute:

(a): The same fuzes are to be subjected to this test: However, if the test within 90 second requirement cannot be maintained, each test or any combination of these tests may be performed on a separate group of fuzes which could increase the maximum sample size for these tests to 256".

Add to Note (b):

"The Fuzing Go and Fuzing No Go impulse shall be imparted to the fuze within 90 seconds of removal from temperature conditioning chamber. All fuzes tested at cold shall be conditioned while sealed in plastic bags."

Add to Notes block:

"(c) Fuzing No Go - 1/2 test sample per temperature; Fuzing Go - 1/2 test sample per temperature.

(d) Selected from fuzes marked (a).

(e) 100% applies only if spherical (burnished) tip configuration is used. A sample of 32 units is required and note (b) applies if flat tip configuration is used.

(f) Prior to performing tests, condition fuzes for two hours after stabilization at indicated temperatures: $-50^{\circ}F \pm 5^{\circ}F$ - 32 fuzes, $+160^{\circ}F \pm 5^{\circ}F$ - 32 fuzes."

Page 74

4.4.2.51: Delete "M230 All Ways Fuze Assembly" and substitute "Fuze, General Purpose Grenade: M230".

Delete Criticals 1 and 2 in their entirety.

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Add Critical 9, 10, 11, 12, and 13.

"9	Safety pin missing	100%	3.4	Visual
10	Fuzing No-Go	100%(e)	3.5.10	4.5.15
11	Arming pin/slider lock	100%	3.8	
12	Presence of Shear Wire	100%	3.4	Visual
13	Auto Trigger	100%	3.9	4.5.15"

Delete "Transportation Vibration 12 3.5.3 4.5.8/4.5.15 4.5.16" and substitute "Transportation Vibration 12 3.5.3 4.5.8/4.5.15 (c)/4.5.16 (c).

Delete "Helicopter Vibration 12 3.5.4 4.5.9/4.5.15/4.5.16" and substitute "Helicopter Vibration 12 3.5.4 4.5.9/4.5.15 (c)/4.5.16 (c)".

Delete "Thermal Shock 12 3.5.6 4.5.11/4.5.15/4.5.16" and substitute "Thermal Shock 12 3.5.6 4.5.11/4.5.15 (c)/4.5.16 (c)".

Delete "Shear Wire Pull Test 32 (a) 3.5.7 4.5.12" and substitute "Shear Wire Pull Test 64 (a) 3.5.7 4.5.12(f)".

Delete "Safety Pin Pull Test 32 (a) 3.5.7 4.5.12" and substitute "Safety Pin Pull Test 64 (a) 3.5.7 4.5.12(f)".

Delete "Arming time 32 (a) 3.5.9 4.5.14 (b)" and substitute "Arming time 62 (a) 3.5.9 4.5.14 (f)".

Delete major 104 in its entirety.

Major 105, delete "(a)" and substitute "(d)".

Delete note (a) in its entirety and substitute:

"(a). The same fuzes are to be subjected to this test. However, if the test within 90 second requirement cannot be maintained, each test or any combination of these tests may be performed on a separate group of fuzes which could increase the maximum sample size to these tests to 256."

Add to Note (b):

"The Fuzing Go and Fuzing No Go impulse shall be imparted to the fuze within 90 seconds of removal from temperature conditioning chamber. All fuzes tested at cold shall be conditioned while sealed in plastic bags."

Add to Notes block:

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"(c) Fuzing No Go - 1/2 test sample per temperature; Fuzing Go - 1/2 test sample per temperature.

(d) Selected from fuzes marked (a).

(e) 100% applies only if spherical (burnished) tip configuration is used. A sample of 32 units is required and note (b) applies if flat tip configuration is used.

(f) Prior to performing tests, condition fuzes for two hours after stabilization at indicated temperatures: -50°F ± 5°F - 32 fuzes, +160°F ± 5°F - 32 fuzes."

Add Majors 107,108, 109, and 110 as follows.

"107	Torque requirement	115	3.4	Gage
108	Wrong adhesive	100%	3.4	Certification
109	Improper cure of adhesive (time and condition)	100%	3.4	Visual/Charts
110	Excessive adhesive (internal)	15	3.4	4.5.18"

4.4.2.52, Add include Page (74a).

4.4.2.53, Add include Page (74b).

Page 75

4.4.4, Delete entire last sentence "The contractor shall submit ... fabrication of such equipment."

4.5.1, Delete in its entirety and substitute:

"4.5.1 Porosity check. A random sample shall be chosen from each cavity of an incoming lot of Fuze Body die cast blanks in accordance with the requirements of Table II of QQ-591. The castings shall then be machined to final configuration and inspected for size, location, depth and frequency of surface defects. The acceptable limit of defects is described in both Table II (attached) and Figure 1 (attached). If the number of rejects from a given cavity exceeds the AQL Level reported in Table II of QQ-591, the incoming quantity from that cavity may be either be rejected and removed from production or screened 100% after machining. A cavity, once pulled from production, cannot be used until requalified. Sectioning may be performed on any portion of the inspection sample if necessary. The firing pin bore shall be inspected 100% by using an Otoscope to direct light and magnify the image."

Add new Figure 1 included (75a).

Add new Table II included (75b).

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.52	Screw, Modified Drive		9390620
			NEXT HIGHER ASSEMBLY: 9333825 or 9345168

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major 101	Thread Body Diameter		0.40%	3.4	Gage
Minor 201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.53	Barrier Plate Assembly with Retainer Tape		12561493
			NEXT HIGHER ASSEMBLY 9333825

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major 101	Tape properly secured		100%	3.4	Visual

NOTES:

Replaces DRDAR-QA Form 160a, 1 Jun 83, which may not be used.

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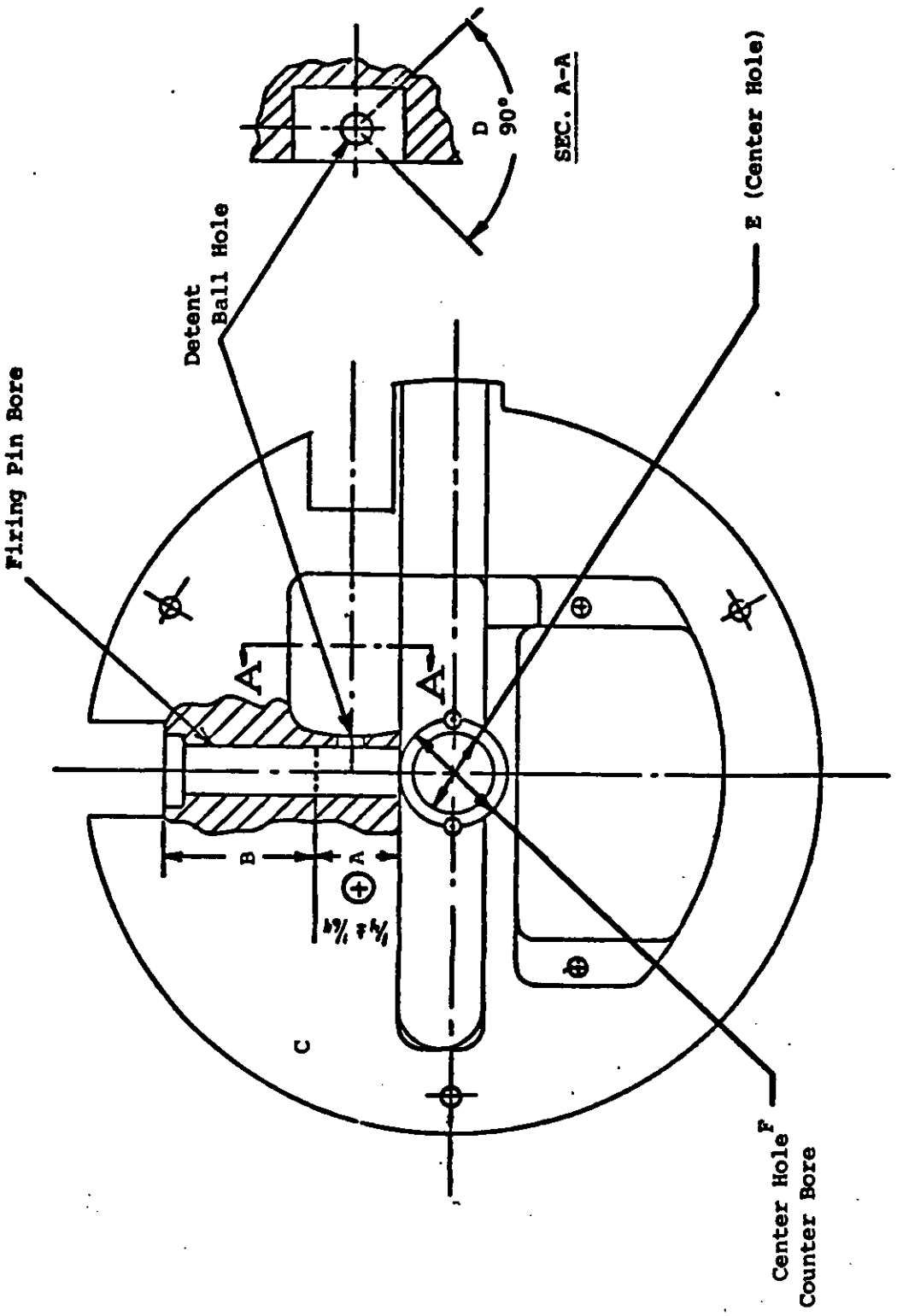


FIGURE 1. Defect Zones

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TABLE II

Maximum Allowable Pore Size and Frequency per Location

ZONE A

Size	Frequency
.040-.060	1
.010-.040	see note 1
.000-.010	no limit

Zone B

Size	Frequency
.040-.100	1
.010-.040	see note 1
.000-.010	no limit

ZONE C

Size	Frequency
.010-.040	see note 1
.000-.010	no limit

ZONE D

Size	Frequency
.000-.010	4 see note 2

ZONE E

Size	Frequency
.040-.060	2
.010-.040	see note 1
.000-.010	no limit

ZONE F

Size	Frequency
.000-.010	4

NOTE 1: Surface defects having a maximum linear dimension of up to .040 are allowed throughout the casting (ZONE A, B, C, D AND E) except in the detent ball hole (ZONE D) and the center hole counter bore (ZONE F). A sum total of 50 is allowed through ZONES A, B, C, and E.

NOTE 2: Permissible defects in the detent ball hole must not extend into the edges of the hole.

DEFINITIONS:

ZONE A - The length of the firing pin bore (.158"+.003" in diameter) extending back from the slider slot a distance of 1/4" ± 1/64" (see figure 1).

ZONE B - The length of the firing pin bore extending back from the end of ZONE A to and including all surfaces of the firing pin cap counterbore (.218"+ .003" diameter). (see figure 1).

ZONE C - All other surfaces of the casting excluding the detent ball hole (ZONE D) and the center hole counter bore (ZONE F).

ZONE D - The surface of the detent ball hole included within an arc of 90° the center of which is the nominal contact point of the detent ball (see figure 1).

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4.5.2, In lines 2 and 3 delete "in the case of a coil spring, or bent in the case of a flat spring".

Page 76

4.5.3. Delete in its entirety and substitute:

"4.5.3 Spring Set. The slider spring and firing pin spring shall be assembled in a restraining fixture in accordance with 4.4.4 and compressed to a nominal height of .430 inch. The springs in their fixtures shall then be conditioned at 175°F ± 5°F for 24 hours. After removal from the oven, the springs shall be allowed to cool to ambient temperature for 2 hours prior to measurement of load. Any spring that fails to maintain load requirement as specified in the appropriate drawings shall be classified defective."

4.5.4, In line 7 delete "immediately before the detonator lot is assembled into the sliders".

4.5.5, In lines 8 and 9 delete "above and within .010 inch of" and substitute ".04 ± .01 inch above".

In lines 13 and 14 delete "The diameter shall measure a minimum of .125 inch and".

Page 78

4.5.8, In third paragraph delete last sentence "The test shall be performed on the fuzes while still mounted in the vibration fuze holding fixtures".

Delete last paragraph in its entirety "The remaining fuzes ... from the holding fixtures" and substitute "The remaining fuzes tested at hot and the remaining fuzes tested at cold shall be subjected to the Fuzing Go Test Fuzes that fail to function shall be considered operability failures. See 4.5.16 for the quantity of Fuzing Go failures that will reject the lot".

4.5.9. Delete "MIL-STD-810C" and substitute "MIL-STD-810B".

PAGE 80

4.5.11. Delete "The fuzes shall not require special preparation prior to this test" and substitute "Each fuze shall be sealed in a temperature stable plastic bag prior to the test."

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Delete the last sentence "A single operability failure shall reject the lot" and substitute "A single Fuzing No Go failure shall reject the lot: See 4.5.16 for the quantity of Fuzing Go failures that will reject the lot".

4.5.12. Delete "less than 16.5 pounds or more than 19.5 pounds" and substitute "less than 16 pounds or more than 22 pounds".

Add to the end of the first paragraph "The load applying device must cause axial displacement of the arming pin at a constant rate of .040 ± .030 inches per second."

Delete last paragraph "After the test, cut shear wires may be replaced and the fuzes returned to the production lot."

PAGE 81

4.5.13. Delete "less than 1 pound or greater than 2 pounds" and substitute "less than 1 pound or greater than 8 pounds".

Add to end of second paragraph "The load applying device must cause axial retraction of the safety pin from the fuze at a constant rate of .040 ± .030 inches per second."

Delete "One failure shall reject the lot" and substitute "Two failures shall reject the lot".

4.5.14, Delete the last two sentences of the second paragraph and substitute "A single reject below minimum arming time shall reject the production lot. Two rejects above maximum time or two jam-ups shall reject the production lot. One jam-up shall result in a retest of 64 fuzes (32 hot, 32 cold). One jam-up during the retest shall reject the production lot. Two rejects above maximum time or one jam-up shall reject a First Article Sample.

ANY TIME IN EXCESS OF 3 SECONDS SHALL BE CONSIDERED A JAM-UP".

End of first paragraph delete "1.75 ± .25 seconds" and substitute 1.50 ± .50 seconds".

4.5.15, Delete in its entirety and substitute:

"4.5.15 Auto trigger, fuzing no-go and arming pin/slider lock. Each fuze shall be subjected to and pass the Auto trigger, Fuzing No-Go and the Arming Pin/Slider Lock Test.

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WARNING: This test is dangerous and all necessary safety precautions shall be taken. Test operators shall be adequately shielded from the fuzes and their holding fixtures during arming, application of impulse, and re-safing.

The fuze shall be secured in a holding fixture in such manner that the shock impulse that the fuze senses will not be attenuated or magnified beyond that applied. The design of the fixture shall be such as to not support the fuze beneath the barrier plate except in the area of the fuze seat.

After mounting the fuze to the holding fixture, but prior to performing the test, the safety pin and shear wire shall be removed. Removal of either the safety pin or shear wire shall be considered as beginning of the arming operation and thus safety shielding requirement applies.

Upon removal of the safety pin, observe that the slider remains in the fully safe position. Should the fuze arm upon removal of the safety pin, it shall be rejected as a critical failure and forced to function before removal from the fixture in a manner as described in the end of this test description.

If the slider remains in the safe position upon removal of the safety pin, shear wire shall be removed. The fuze shall then be allowed to arm by raising the arming pin. The trigger shall not spontaneously release the detent ball and firing pin when the slider moves to the fully armed position. Any fuze that fails this requirement shall be classified as a critical failure. If the Fuzing No-Go test is not required, skip to the Arming Pin/Slider Lock Test. Otherwise, proceed to the Fuzing No-Go test.

With the fuze in the armed state, an impulse shall be applied to the fuze/fixture in a direction parallel to the axis of the fuze within one degree. The magnitude of the impulse shall be 180 + 20 gravities. Duration of the impulse shall be 1.5 $\pm$ 0.5 milliseconds. The orientation of the fuze with respect to the impulse shall be such as to simulate a normal target impact.

The trigger shall not release the detent ball and firing pin as a result of the shock impulse. Any fuze that functions during the No-Go test shall be classified as a critical failure.

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Fuzes that pass the No-Go test shall be subjected to the arming Pin/Slider Lock Test by moving the slider back into its fully safe position and lowering the arming pin until it is fully seated. A new shear wire shall be installed. At this time observe that the slider remains in the fully safe position when unobstructed. Should arming occur, the fuze shall be rejected as a critical failure and forced to function before removal from the fixture in a manner as described in the end of this test description.

Fuzes that pass the Arming Pin/Slider Lock Test shall be resafed by reinsertion of the safety pin. Resafing is considered complete at the end of installation of both the shear wire and safety pin.

Any fuze that cannot be properly resafed shall be rearmed in the fixture and forced to function before removal from that fixture. This can be accomplished by increasing the magnitude of the impulse to approximately 1,000 gravities. Should the armed fuze still not trigger, a hang-fire has occurred and must be handled accordingly. Provision must be made that will allow for removal of the failed fuze from the test equipment while maintaining a shielded condition."

Page 82

4.5.16, Delete in its entirety and substitute.

"4.5.16 Fuzing go test. A random sample of fuzes shall be selected from each production lot and subjected to the Fuzing Go Test.

WARNING: This test is dangerous and all necessary safety precautions shall be taken. Test operators shall be adequately shielded from the fuzes and their holding fixtures during arming,application of impulse, and re-safing.

The fuze shall be secured in a holding fixture in such a manner such that the shock impulse that the fuze senses will not be attenuated or magnified beyond that applied. The design of the fixture shall be such as to not support the fuze beneath the barrier plate except in the area of the fuze seat.

After mounting the fuze to the holding fixture, but prior to performing the test, the safety pin and shear wire shall be removed. Removal of either the safety pin or shear wire shall be considered as beginning of the arming operation and thus safety shielding requirement applies.

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Upon removal of the safety pin, observe that the slider remains in the fully safe (out-of-line) position. Should the fuze arm upon removal of the safety pin, the fuze shall be resafed and rejected as a critical arming failure.

If the slider remains in the safe position upon removal of the safety pin, the shear wire shall be removed. The fuze shall then be allowed to arm by raising the arming pin to the end of its stroke at which time the spring lock will engage.

With the fuze in the armed state, an impulse shall be applied to the fuze/fixture in a direction parallel to the axis of the fuze within one degree. The magnitude of the impulse shall be 700 gravities maximum. Duration of the impulse shall be 1.0 $\pm$ 0.5 milliseconds. The orientation of the fuze with respect to the impulse shall be such as to simulate a normal target impact.

The trigger shall release the detent ball and firing pin causing initiation of the explosive train. Any fuze that fails to function shall be considered a failure. A sum total (for all fuzing Go Test) of three failures shall reject the production lot. Two failures shall result in a retest (including transportation vibration, Thermal shock, Helicopter Vibration, and Temperature Conditioning) of 82 fuzes. One failure during the retest shall reject the production lot. Two failures shall reject the First Article Sample.

Any fuze that fails the Fuzing Go Test shall be forced to function before removal from the fixture. This can be accomplished by increasing the magnitude of the impulse to approximately 1,000 gravities. Should the armed fuze still not trigger, a hang-fire has occurred and must be handled accordingly. Provision must be made that will allow for removal of the failed fuze from the test equipment while maintaining a shielded condition."

Pages 21, 22, 26, 27, 34, 35, 36, 37, 38, 62, and 63

Add: "9333825/9345168" in Next Higher Assembly block.

Pages 23,24, 25 and 29

Add: "9347185" in Next Higher Assembly block.

Page 83

Add new 4.5.18 as follows:

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"4.5.18 Lead pull test. A random sample of Barrier Plate Assembly (M230) shall be subjected to this test. The lead shall withstand an axial unseating load of 2.0 lbs. min. without being pushed out of barrier plate. One or more failures to meet this requirement shall cause the lot to be rejected. Warning. This test is considered dangerous. Test operators shall be adequately shielded during testing."

Add new 4.5.19 as follows:

"4.5.19 Excessive adhesive (internal). A random sample of fifteen (15) Fuzes shall be inspected internally for excessive adhesive by removal of the barrier plate.

WARNING: This inspection is dangerous and all necessary safety precautions shall be taken. Inspectors/operators shall be adequately shielded from the fuze and their holding fixtures during disassembly and inspection, in accordance with DARCOM-R 385-100, paragraphs 25-3 and 25-4.

Prior to disassembly, the fuze shall be inspected to insure the shear wire and safety pin are in place. A firing pin assembly clip shall also be applied to end of the firing pin body groove to prevent inadvertent movement of the firing pin.

The fuze shall be supported in a fixture during disassembly and inspection. The fixture shall contain a feature which will prevent inadvertent arming of the slider. The fuze shall be disassembled by removing the three drive screws and the support screw. After the said screws are out, remove the barrier plate and inspect the interior for evidence of adhesive. Adhesive in the pivot screw/fuze body interface is permitted as long as it does not extend into the cavity of the pivot screw or into the cage of the fuze body. Any evidence of the adhesive in the interior of the fuze, except for the areas noted above, shall constitute a defective fuze. One or more defective fuze shall cause the lot to be rejected.

All fuzes subjected to this inspection are considered destroyed and shall be disposed of in accordance with in-house procedures".

5.2, Delete "a 3 inch X 3 inch zip-lock type bag" and substitute "plastic bag".

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